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SUPERSEDING
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November 30, 1961

FEDERAL SPECIFICATION

SEWING, MACHINE, HOUSEHOLD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sewing machines for domestic use.

1.2 Classification.

1.2.1 *Classes and styles.* Sewing machines covered by this specification shall be of the following classes and styles, as specified (see 6.2):

Class 2—Motor driven, portable model, full size head.
Style E—With round bobbin, rotary sewing mechanism and reverse stitch.
Class 4—Motor driven, cabinet table model, full size head.
Style C—With round bobbin, oscillating sewing mechanism, and reverse stitch.
Style F—With round bobbin, rotary sewing mechanism, zigzag and reverse stitch.
Class 5—Foot operated, cabinet table model, full size head.
Style C—With round bobbin, oscillating sewing mechanism, and reverse stitch.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CC–M–636—Motor, Alternating-Current, (Fractional Horsepower).
KK–B–211—Belting; Round, Leather.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Service Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.
MIL–L–10547—Liner, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

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MIL–I–16910—Interference Measurement, Electromagnetic, Methods and Limits.

MIL–E–55301—Electromagnetic Compatibility.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–130—Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association (NFPA) Publication:

NFPA 70–National Electrical Code.

(Copies may be obtained from the National Fire Protection Association, 60 Batterymarch St., Boston, Mass., 02110.)

Underwriters' Laboratories (UL), Inc. Standard:

No. 73—Motor-Operated Appliances.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, N. Y., 11749; 207 E. Ohio St., Chicago, Ill., 60611; or 1655 Scott Blvd., Santa Clara, Calif., 95050.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample sewing machine covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for inspection and approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacturer of the production items.

3.2 Materials. The materials used shall conform to the applicable specifications and requirements specified hereinafter. When materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice.

3.2.1 *Belts.* All leather belting furnished shall conform to the requirements of KK–B–211. Rubber or composition belts shall be in accordance with standard commercial practice.

3.2.2 *Motors.* All fractional horsepower, single phase, universal type motors furnished shall conform to the requirements of CC–M–636.

3.3 Standard product. Machine and equipment shall be new, of the manufacturer's current production, complete with all necessary parts and equipment, self-contained, rugged in construction, made of unused parts, present a neat appearance, and shall be ready for sewing. Moving parts shall be adequately balanced as necessary to prevent objectionable vibration of machine, when operated on a level floor for cabine[illegible]ble machines or on a level table for p[illegible] machines.

3.4 Design. Machines shall be suitable for stitching dresses, blouses, underclothing, nightgowns, pajamas, shirts, lightweight clothing, and other operations on articles made of light-or-medium-weight fabrics.

3.4.1 All stitch and seam types referred to in this specification shall conform to the requirements of Fed. Std. No. 751.

3.4.2 *Safety standards.* All electrical equipment shall conform to the UL standards for safety as applicable.

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3.4.3 *Certification.*

3.4.3.1 Prior to approval of preproduction sample, if one is submitted, or prior to approval of the first shipment of sewing machines, the contractor shall submit to the contracting agency proof that the sewing machines he proposes to supply under this specification conform to the requirements of the UL, Inc. Standard No. 73. The label, or listing with reexamination, of the UL, Inc. may be accepted as evidence that the sewing machines conform to this requirement.

3.4.3.2 In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the sewing machines conform to the requirements, including methods of test, of the UL, Inc. Standard No. 73.

3.4.3.3 Compliance with the above preliminary requirements for certification does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his sewing machines.

3.4.4 *Electromagnetic compatibility.*

3.4.4.1 *Army purchases.* Unless otherwise specified (see 6.2), sewing machines procured under this specification for Army usage, shall conform to the administrative requirements of MIL–E–55301 (see 4.4.2.1).

3.4.4.2 *Navy and Air Force purchases.* When specified (see 6.2), equipment procured under this specification for Navy and Air Force use shall conform to MIL–I–16910 relative to radiated and conducted radio interference. The paragraph in MIL–I–16910 which excludes engine generators and miscellaneous engine-driven equipment having power output ratings in excess of 60 kilowatts from the requirements for radio interference shall not apply. Radio-interference suppression and tests therefore, will not be required for electric motors without commutation and having not more than one starting contact and operated at 3600 or less r.p.m. (see 4.4.2.2).

3.4.5 *Electric light fixture.* All motor-operated machines shall have a light fixture, either attached to or be an integral part of the machine, and shall be suitable for illuminating the area of the bed at the needle without interfering with the handling of the fabric or the vision of the operator. A lamp rated at 6 watts or more and a suitable shade and switch shall be furnished.

3.4.6 *Bobbin winder.* A device to wind the bobbins shall be provided. The device shall be actuated without separate power by friction contact with the balance wheel of machine, by drive belts, or may be integral with the sewing mechanism. On round bobbin machines, the bobbin winder shall have a suitable tension arrangement for controlling the thread to wind thread evenly and with proper tension on the bobbin. The friction contact shall be made manually with arrangements to hold contact until automatically released when bobbin becomes full, and also released manually before bobbin becomes full, as desired. When the bobbin winding is integral with the sewing mechanism, a button or lever shall be provided to manually or automatically start and stop the winding operation. A separate spool pin with the necessary thread guides may be provided for each machine to hold the spool of bobbin thread to permit winding bobbins.

3.4.7 *Motor.* A motor shall be furnished for all motor-operated machines and shall be universal type, suitable for operation on alternating current frequencies from 50 to 60 cycles, and on direct current circuits. Unless otherwise specified (see 6.2), the operating voltage shall be 120 volt. The motor shall be suitably mounted on or in the head of the machine and may be friction drive mechanism, belt drive, or direct-gear drive.

3.4.7.1 *Friction drive.* When friction drive is furnished, motor shall be mounted to machine head in such a manner that a pulley on motor shaft will contact the rim of balance wheel on machine. Motor pulley shall have a metal core and either a cork or rubber outer friction surface. An adjustable spring arrangement shall be provided to hold

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pulley against balance wheel for proper friction contact.

3.4.7.2 *Belt drive.* When belt drive is furnished, motor shall be mounted to machine head with the motor pulley in alignment with the groove in balance wheel of machine. A belt specified in 3.2.1 shall be furnished between the motor and the balance wheel. Adjustment shall be provided to maintain proper belt tension.

3.4.7.3 *Direct-gear drive.* When direct-gear drive is furnished, motor shall be mounted in such a manner that the gear on motor shaft will align with a gear on arm shaft or hub of balance wheel of machine.

3.4.8 *Lubrication.* Means shall be provided to readily and properly apply lubrication to all bearings and frictional surfaces on machine, motor, and equipment. Gears that are required to be packed with grease shall be suitably covered to retain grease and to exclude dust and lint.

3.4.9 *Marking for identification.* Each machine, controller, and external mounted motor shall be permanently and legibly identified with the class specified herein, and with the manufacturers' name or trademark. Marking shall be in accordance with MIL–STD–130. Serial numbers may be indented in head or bed of machine.

3.4.10 *Tools and accessories.* Tools and accessories normally furnished shall be provided with each machine.

3.4.11 *Tension.* A thread arrangement shall be provided with each machine for each thread used by the machine and shall be adjustable to the amount of tension required to sew satisfactory stitches. The machine shall be equipped with an automatic upper thread release, effective when the presser foot is raised to the maximum height. An auxiliary thread takeup spring, adjustable for tension and position, shall be provided for the needle thread, except when a rotary thread takeup arrangement is furnished, an auxiliary takeup spring without adjustment may be provided. Guides for thread shall be smooth to prevent fraying of thread.

3.5 Wiring. All motor-operated machines shall be completely wired, ready for operation, with an extension cord of flexible cable of sufficient length to extend beyond the machine for a minimum distance of six feet. The cord shall be equipped with, and connected to, an attachment plug having a finger grip. Wiring shall comply with the latest published regulations of the National Electrical Code and all connections shall be made between motors, controllers, lights installed on the machine, and the extension cord.

3.5.1 The electrical equipment shall be suitable for the voltage and current specified in the invitation for bids and shall conform to the latest standards of the National Electrical Manufacturers' Association and shall conform to UL, Inc., where applicable.

3.6 Construction. Machines shall be single-needle design, equipped with a sewing mechanism for producing 2-thread lock-stitch (type 301) and zigzag stitch (type 304), as specified (see 1.2.1 and 6.2); shall be round bobbin, oscillating sewing mechanism; or round bobbin, rotary sewing mechanism, as specified for each class and style of machine. The sewing mechanism shall conform to the requirements specified herein.

3.6.1 *Round bobbin, oscillating sewing mechanism.* When round bobbin, oscillating sewing mechanism is furnished, machine shall be equipped with an oscillating shuttle or hook, which will operate back and forth (oscillate) in either a vertical plane on a horizontal axis, or in a horizontal plane on a vertical axis. A bobbin case or holder, having an adjustable tension arrangement, shall be provided to hold the bobbin. On the horizontal axis design, the bobbin or bobbin case shall be inserted from under the bed, and on vertical axis design, the bobbin case shall be inserted from the top of the bed. Unless otherwise specified herein, bobbin shall have a capacity of approximately 60 yards of No. 60 thread and shall be secured in operating position so bobbin or bobbin case will not fall out while the machine is in operation or when being moved, and will remain in place

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until removed or released manually. Machine shall be fitted with either a cam-driven or link-actuated thread takeup lever for the needle thread. When bobbin winding operation is built internally with the sewing mechanisms, thread capacity of the bobbin shall be not less than 20 yards.

3.6.2 *Round bobbin, rotary sewing mechanism.* When round bobbin, rotary sewing mechanism is furnished, machine shall be equipped with a rotary hook or shuttle, operating in a full rotary motion either in a vertical plane on a horizontal axis, or in a horizontal plane on a vertical axis. Rotary motion of hook may be at the ratio of 1, 2, or 3 revolutions to each needle bar stroke. A bobbin case or holder having an adjustable tension arrangement shall be provided to hold the bobbin. On the horizontal axis design, the bobbin or bobbin case shall be inserted from under the bed, and on vertical axis design, the bobbin or bobbin case shall be inserted from the top of the bed. Unless otherwise specified herein, bobbin shall have a capacity of approximately 60 yards of No. 60 thread and shall be secured in operating position so that bobbin or bobbin case will not fall out while the machine is in operation or when being moved, and remain in place until removed or released manually. Machine shall be fitted with either a cam-driven, a link-actuated thread takeup lever, or a rotary thread takeup arrangement for the needle thread. When the bobbin winding operation is built internally with the sewing mechanism, thread capacity of the bobbins shall be not less than 20 yards.

3.6.3 *Feed mechanism.* Machine shall be equipped with a positive, 4-motion, drop feed, whereby the serrated surface of the feed first raises above the level of the throat or needle plate, contacting the work, then moves the amount of the desired length of stitch carrying the work with it. After the feeding movement is completed, the feed surface drops below the level of the feed plate and returns to its original position, completing its cycle. Mechanism controlling the four motions shall be positive, except that a spring may be used to assist the lowering action of the feed. A hinged- or walking-type presser foot shall be furnished. It shall be mounted on a suitable bar or holder allowing the foot to raise or lower by the action of the feed. It shall also be possible to raise or lower the foot manually for the purpose of inserting, moving, or removing the fabric being sewed. An adjustable spring pressure arrangement shall be provided for the presser foot to give the proper amount of pressure required on the fabric being sewed. The feeding motion shall be accomplished while the needle of the machine is out of the fabric being sewed. The size and type of feed, feed plate, and presser foot shall be suitable for the purpose intended. A minimum clearance or capacity of 1/4-inch between the presser foot in the "up" position and the needle plate shall be provided. Arrangement shall provide for adjusting the height of the feed dog so that the serrated surface of the feed may be raised above or below the level of the throat plate.

3.6.3.1 *Reverse stitch mechanism.* The machine shall be equipped with the positive 4-motion, drop feed complying with the requirements of 3.6.3 and, in addition, shall be designed to reverse the feeding action of the drop feed for backtack stitching, as desired. The feeding direction shall be controlled by the stitch regulator, or a separate lever for the purpose, and shall be capable of being changed while the machine is in operation.

3.6.3.2 *Zigzag (swing-needle) stitch.* When specified that machine be of the zigzag or swing-needle stitch style with reverse mechansim, machine shall be equipped with the positive 4-motion, drop feed complying with the requirements of 3.6.3. In addition to straight-stitch sewing, the machine shall be equipped with control(s) for swinging the needle bar from side to side of the sewing line or to the right or left to produce a zigzag stitch that will satisfactorily perform decorative stitching, embroidery, make buttonholes and sew on buttons. The reverse

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mechanism shall be in accordance with the requirements of 3.6.3.1.

3.6.3.3 *Stitch regulator.* A stitch regulator, conveniently located, capable of adjusting the length of stitch over a range from 7 to 30 stitches per inch, shall be provided for each machine. Marking shall be provided to indicate the number of stitches required.

3.6.4 *Tabletop, base, etc.* Unless otherwise specified herein, tabletop base, and other parts that are specified to be made of wood, with the exception of legs, stretchers, runners, and rails, shall be solid wood construction of walnut, mahogany, oak, maple, magnolia, birch, or gum; or laminated wood construction having a hardwood center core and at least a 2-ply veneer, not less than 13/16-inch finished thickness, of walnut, mahogany, oak, maple, magnolia, birch, or gum for the exposed or exterior surfaces. Legs, stretchers, runners, and rails shall be solid wood construction of the kind of wood used for the enclosing cabinet; or of gum wood construction stained to match the exterior surfaces of the enclosing cabinet. The exterior surfaces of the tabletop, base, enclosing cabinet, legs, etc., shall be stained, filled, sealed, and finished in the natural grain of the wood in accordance with industry practice. Tabletop shall be securely fastened to supporting stand or cabinet.

3.6.4.1 Dust trays, where required under machine, shall be of at least 3-ply construction hardwood, steel, or pressed wood construction.

3.6.5 *Controller.* All motor-operated machines shall have a foot- or knee-operated sewing speed controller. The controller shall be enclosed and provide gradual change of sewing speeds without jerking or sudden speed jumps.

3.6.5.1 *Foot control.* When foot control is furnished, rubber pad or pads shall be provided on base to hold control to position set on floor.

3.6.5.2 *Knee control.* When knee control is furnished, it shall be fastened in a suitable position for knee-height operation. On cabinet model machine, a hinged-knee lever shall be provided which will allow lever to be moved out of sight when machine is not in use. On portable model machine, a knee lever, which may be removed when machine is not in use, or a foot control shall be provided.

3.6.6 *Dimensions of machine heads.* Each machine head shall comply with the requirements for dimensions as specified in table I.

Table I. *Dimensions*

Class	Bed		Distance from needle to inner side of base of vertical arm inches, (minimum)
	Length, inches (minimum)	Width, inches (minimum)	
2	13	6–3/4	6–5/8
4	13	6–3/4	6–5/8
5	13	6–3/4	6–5/8

3.6.7 *Class 2.* Motor driven, portable model, full size head.

3.6.7.1 *Style E.* With round bobbin, rotary sewing mechanism and reverse stitch.

3.6.7.1.1 *Dimensions and working space.* The dimensions of the head and the working space shall comply with the requirements for class 2 machine, as specified in table I.

3.6.7.1.2 *Sewing mechanism.* Sewing mechanism shall be round bobbin, rotary sewing mechanism conforming to the requirements of 3.6.3.

3.6.7.1.3 *Sewing speed.* Machine shall be capable of operating up to a speed of 1,200 stitches per minute.

3.6.7.1.4 *Base.* Machine shall be mounted on a wooden or plastic base, cut out to allow the bedplate of the machine to be flush with the top of the base.

3.6.7.1.5 *Cover.* A wooden, sheet-metal or plastic cover with a handle at the top or a carrying case made of suitable materials and

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hinged lid with a handle at the top, for carrying the equipment shall be provided. Cover or lid shall be capable of being securely fastened or locked to the base or case for carrying purposes. The base and cover or the case shall also provide storage for equipment.

3.6.8 *Class 4.* Motor driven, cabinet table model, full size head.

3.6.8.1 *Style C.* With round bobbin, oscillating sewing mechanism and reverse stitch.

3.6.8.1.1 *Dimensions and working space.* The dimensions of the head and the working space shall conform to the requirements for class 4 machine, as specified in table I.

3.6.8.1.2 *Sewing mechanism.* Sewing mechanism shall be round bobbin, oscillating sewing mechanism, conforming to the requirements of 3.6.1 and, in addition, a reverse stitch mechanism, conforming to the requirements of 3.6.3.1, shall be provided.

3.6.8.1.3 *Sewing speed.* Machine shall be capable of operating up to a speed of 1,000 stitches per minute.

3.6.8.1.4 *Cabinet table.* Machine shall have a tabletop, extension leaf or leaves, and cabinet frame made of wood in the species specified with the legs made of wood or metal stand having a foot treadle. All parts made of wood shall conform to the requirements of 3.6.4. Tabletop shall be cut out to fit the machine bed, and in such a manner that when machine is in operating position, its bed will be flush with the tabletop, with felt, rubber, or leather pads between bed of machine and recessed tabletop to eliminate objectionable noise or vibration. Hinges shall be provided so machine can be tilted back for oiling and cleaning, together with means whereby machine may be lowered into the cabinet and extension leaf or leaves folded over the opening. The tabletop shall be not less than 15 inches wide; with the extension leaf or leaves closed, top shall be not less than 21 inches long; with the extension leaf or leaves open, top shall be not less than 36 inches long. The height of the cabinet table shall be approximately 30 inches. No less than one drawer, to accommodate attachments and accessories, shall be furnished. A shield or dust tray, mounted in a manner to permit the head to be lowered into cabinet, shall be provided under bed of machine.

3.6.8.2 *Style F.* With round bobbin, rotary sewing mechanism, zigzag and reverse stitch.

3.6.8.2.1 Requirements of 3.6.8.1.1 and 3.6.8.1.4 shall apply and, in addition, it shall be equipped with a zigzag stitch mechanism in accordance with 3.6.3.2 and a reverse stitch mechanism conforming to the requirements of 3.6.3.1.

3.6.8.2.2 *Sewing mechanism.* Sewing mechanism shall be round bobbin, rotary sewing mechanism conforming to the requirements of 3.6.2.

3.6.8.2.3 *Sewing speed.* Machine shall be capable of operating up to a speed of 1,200 stitches per minute.

3.6.9 *Class 5.* Foot-operated, cabinet table model, full size head.

3.6.9.1 *Style C.* With round bobbin, oscillating sewing mechanism and reverse stitch.

3.6.9.1.1 *Dimensions and working space.* The dimensions of the head and working space shall comply with the requirements for class 5 machines as specified in table I.

3.6.9.1.2 *Sewing mechanism.* Sewing mechanism shall be round bobbin, oscillating sewing mechanism conforming to the requirements of 3.6.1.

3.6.9.1.3 *Supporting stand and cabinet table.* Machine shall have a tabletop extension leaf or leaves, supporting stand, and head enclosing cabinet. The tabletop and leaf or leaves shall be of wood conforming to the requirements of 3.6.4, and the top shall be cut out to fit the machine in such a manner that when machine is in operating position, its bed will be flush with the tabletop, with felt, rubber, or leather pads between the bed and recessed table to eliminate objec-

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tionable noise or vibration. Suitable hinges shall be provided so machine can be tilted back for oiling and cleaning, together with suitable means whereby machine may be lowered into cabinet and extension leaf or leaves folded over the opening. The table top shall be not less than 15 inches wide; with the extension leaf closed, top shall be 38 inches long; with the extension leaf open, top shall be 46 inches long. Supporting stand shall be metal, and together with the table top, shall be approximately 30 inches high. The table or stand shall have no less than three drawers. A shield or dust tray, mounted in a manner to permit the head to be lowered into the cabinet, shall be provided under the bed of the machine.

3.6.9.1.4 *Foot power drive.* Machine shall be driven by a belt from a band wheel actuated by a foot treadle through a pitman rod. The pitman shall have a ball and socket joint at the treadle end and a ball bearing at the crank end. Band wheel and crank shaft shall be mounted between trunnion bearings. Treadle shall be of iron or steel supported between trunnion bearings, and if of the grate-opening type, the openings shall not exceed 1 inch in width. A safety dress shield shall be furnished for the band wheel, together with means to permit throwing the belt either off or on the band wheel, as required.

3.7 Handbook of instructions. The contractor shall furnish one copy of a handbook of instructions with each sewing machine supplied under this specification. The handbook shall contain operating instructions, illustrations, parts list, and maintenance of the sewing machine.

3.8 Workmanship. The design, materials, workmanship, finish operation and details of sewing machines and related equipment shall be suitable and adequate for the purpose intended and shall be free from defects or blemishes which affect the appearance, or which may affect the serviceability or operation. All surfaces that might otherwise interfere with the operation or movement of the fabrics being stitched, shall be smooth and free from sharp edges and projections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects listed in table II, dimensions specified and tested as specified in 4.4.1.

4.3 Sampling for inspection. Sampling for inspection shall be made in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated hereinafter.

4.3.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 *Intermediate visual and dimensional examination.* The intermediate product or assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for compliance with this specification.

4.3.3 *Inspection of the end item.* The lot shall consist of all sewing machines submitted for inspection at one time. The sample unit shall be one sewing machine.

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4.3.3.1 *Visual examination.* Visual examination of the completely assembled end item shall be in accordance with the classification of defects in table II. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

Table II. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Design	Characteristic not in accordance with the specified requirements unless otherwise classified herein	X	
Construction and workmanship	Parts missing, unless otherwise classified herein, or not specified type	X	
	Fractured, split, malformed, or otherwise impaired	X	
	Functioning component that requires abnormal force to operate		X
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended	X	
	Any component not readily accessible for servicing where required		X
Tabletop	Not of specified wood	X	
	Not fastened as specified	X	
Bolts, nuts, screws, studs, and other types of fasteners	Missing, broken, stripped, fractured		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain; or excessive insulation stripped from wiring		X
	Switch not as specified	X	
	Motors not as specified	X	
	Controller not as specified	X	
	Light fixture not as specified		X
Stand	Not of steel construction	X	
Pulleys	Not type specified	X	
Lubrication (where applicable)	Not lubricated where required	X	
	Insufficiently lubricated		X
	Not specified type lubrication fitting	X	
	Lubrication fitting not readily accessible for servicing		X
Identification marking	Missing, incomplete, not legible, not in proper location, not affixed in the prescribed manner		X

4.3.3.2 *Dimensional examination.* Examination shall be made for compliance with dimensions specified herein. Any deviation therefrom shall constitute a defect. The inspection level shall be S–2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.3.3 *Testing of the end item.* The supplier shall perform tests in accordance with 4.4.1 on each machine with no indication of failure. The radio interference suppression test shall be conducted only when specified.

4.3.4 *Code and standards compliance.* The supplier shall furnish the Government with proof of compliance with the requirements of the National Electrical Code, National

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Electrical Manufacturers Association Standards, and U. L. Standards as applicable to 3.4.3.

4.3.5 *Examination of preparation for delivery.* An examination shall be made to determine that preservation, packaging, packing, and markings as required by section 5 are complied with. Defects shall be scored as specified in table III. The inspection level shall be S–2 with an AQL of 6.5 defects per hundred units.

Table III. *Examination of preparation for delivery*

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps; loose strapping, inadequate stapling; bulging or distortion of containers; blocking and cushioning inadequate, improper, or missing; bolts improperly affixed.

4.4 Tests.

4.4.1 *Performance.* Tests shall be made to determine that the machines meet the operational requirements specified herein, peculiar to each class and style of machine furnished.

4.4.2 *Electromagnetic compatibility test.*

4.4.2.1 *Army purchases.* The preproduction model, equipped for reduction of radio interference in accordance with 3.4.4.1, shall be tested as described in MIL-E-55301 by the supplier or his private testing organization. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer, for approval, the electromagnetic compatibility test report required in MIL-E-55301. The test report will include a block diagram of the test setup, a description of the interference reduction system applied and a list of the test equipment utilized as described in MIL-E-55301. Upon approval by the contracting officer, and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production units.

4.4.2.2 *Navy and Air Force purchases.* When required (see 6.2), sampling, inspection, and test procedures shall conform to the requirements of MIL–I–16910. Tests for radiated and conducted interference on the complete equipment assembly shall be conducted on the production test unit and upon such additional units as are specified. For tests on the complete equipment assembly, all testing equipment, instruments, personnel for making the tests, the tests location, which shall be reasonably free from radiated and conducted interference, and other necessary facilities shall be furnished by the contractor.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Machines.* Class 2 sewing machines shall be preserved method II e, and classes 4 and 5 sewing machines preserved method II a or II e of MIL-P-116. Alternatively, classes 4 and 5 sewing machines shall have the cabinet table packaged method III and the machine head preserved method II a or II e of MIL-P-116. Parts of the machine that are free to move or rotate shall be secured to prevent movement or vibration while in transit.

5.1.1.2 *Tools and accessories.* Tools and accessories shall be preserved and packaged in accordance with method I of MIL-P-116. These components shall then be packaged in a corrugated fiberboard box designed to fit the contents in a compact manner.

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5.1.1.3 *Instruction book.* Each instruction book shall be prepared method IC-1 of MIL-P-116 and packaged in the same box as the spare parts, tools, and accessories.

5.1.2 *Level C.* Sewing machines shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets these requirements.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Class 2 machine.* Each machine, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat steel strapping or tape banding shall be in accordance with the appendix of the container specification.

5.2.1.2 *Classes 4 and 5 machines.* Each class 4 or 5 machine, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to style RSC, V3c or V3s of PPP-B-636, except that weight limitation shall not be exceeded; overseas type of PPP-B-601; class II of PPP-B-591; or class 2, style 2 or 4 of PPP-B-621. Contents shall be secured to prevent movement while in transit. Shipping containers containing method III packages (see 5.1.1.1) shall be waterproofed in accordance with the appendix of the applicable container specification or provided with a type I or II, grade C case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 *Level B.*

5.2.2.1 *Class 2 machines.* Unless otherwise specified (see 5.2.2.3), each machine, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification.

5.2.2.2 *Classes 4 and 5 machines.* Unless otherwise specified (see 5.2.2.3), each class 4 or 5 machine and spare parts, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636, except that weight limitation shall not be waived; domestic type, style A or B of PPP-B-601; class I, style A or B of PPP-B-591; or class 1, style 2 or 4 of PPP-B-621. Closure shall be in accordance with the appendix of the applicable container specification.

5.2.2.3 When specified (see 6.3), weather-resistant fiberboard specified in 5.2.1 shall be used for shipping containers conforming to PPP-B-636.

5.2.3 *Level C.* Sewing machines, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Machines covered by this specification are intended to be used only for plain sewing, patching, mending, darning or the making of simple garments.

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The zigzag type, or swing-needle machine is designed to handle a wider range of stitch styles and specialty sewing such as decorative stitching, embroidery, sewing on buttons and making buttonholes, in addition to straight sewing. The class 5, style C machine is intended for use where a source of electrical power is not available.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Class and style machine desired (see 1.2.1).

(*c*) When preproduction sample is required (see 3.1).

(*d*) Operating voltage required if other than specified (see 3.4.7).

(*e*) When electromagnetic compatibility is not required for Army procurement (see 3.4.4.1, 4.4.2.1, and 6.4).

(*f*) When radio-interference suppression is required for Navy and Air Force procurement (see 3.4.4.2, 4.4.2.2, and 6.4).

(*g*) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

(*h*) If weather-resistant fiberboard shall be used for level B pack (see 5.2.2.3).

6.3 Bid should be accompanied by such evidence as will show that the manufacturer has adequate experience in the manufacture of the type of machine specified in the schedule or proposal. A list of users, with the length of time in service, should be included. The Government reserves the right to reject bids when in the opinion of the procuring agency the manufacturer's experience is inadequate.

6.4 Electromagnetic compatibility. The contracting officer should arrange for submission of contractual samples at least 10 days in advance of required submission date of preproduction sample and should arrange for suppression system investigation at either the designated engineering laboratory or at an interference free area in proximity to the prime or subcontractor's plant.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—69

Review activities:

Navy—SA
Air Force—69

User activities:

Army—EL
Navy—MC

Preparing activity:

Army—GL

CIVIL AGENCIES INTEREST:

COM—NBS
GSA—FSS
DC
INT—BIA
VA

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are tob placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Sewing Machine, Household - GG-S-225c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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